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مباحث: یادگیری تقویتی

Q1
Imagine an unknown environments with four states (A, B, C, and X), two actions (← and →). An agent acting in this environment has recorded the following episode:

s	a	s'	r	Q-learning iteration numbers (for part b)
A	→	B	0	1, 10, 19, ...
B	→	C	0	2, 11, 20, ...
C	←	B	0	3, 12, 21, ...
B	←	A	0	4, 13, 22, ...
A	→	B	0	5, 14, 23, ...
B	→	A	0	6, 15, 24, ...
A	→	B	0	7, 16, 25, ...
B	→	C	0	8, 17, 26, ...
C	→	X	1	9, 18, 27, ...

(a)
Consider running model-based reinforcement learning based on the episode above. Calculate the following quantities:

T^(B, →, C) =
R^(C, →, X) =

(b)

Now consider running Q-learning, repeating the above series of transitions in an infinite sequence. Each transition is seen at multiple iterations of Q-learning, with iteration numbers shown in the table above.
After which iteration of Q-learning do the following quantities first become nonzero? (If they always remain zero, write never).

Q(A, →)?
Q(B, ←)?

Q1
Answer:

(a)
T^(B, →, C) = 2/3
R^(C, →, X) = 1

(b)
Q(A, →) gets value in iteration 14
Q(B, ←) gets value in iteration 22

Q2

Here our MDP is about choosing a ride from a set of many rides.
You start off feeling well, getting positive rewards from rides, some larger than others. However, there is some chance of each ride making you sick. If you continue going on rides while sick there is some chance of becoming well again, but you don't enjoy the rides as much, receiving lower rewards (possibly negative).
You have never been to an amusement park before, so you don't know how much reward you will get from each ride (while well or sick). You also don't know how likely you are to get sick on each ride, or how likely you are to become well again. What you do know about the rides is:

Actions / Rides	Type	Wait	Speed
Big Dipper	Rollercoaster	Long	Fast
Wild Mouse	Rollercoaster	Short	Slow
Hair Raiser	Drop tower	Short	Fast
Moon Ranger	Pendulum	Short	Slow
Leave the Park	Leave	Short	Slow

We will formulate this as an MDP with two states, well and sick. Each ride corresponds to an action. The 'Leave the Park' action ends the current run through the MDP. Taking a ride will lead back to the same state with some probability or take you to the other state. We will use a feature based approximation to the Q-values, defined by the following four features and associated weights:

Features	Initial Weights
$f_0(state, action) = 1$ (this is a bias feature that is always 1)	$w_0 = 1$
$f_1(state, action) = \begin{cases} 1 & \text{if } action \text{ type is Rollercoaster} \\ 0 & \text{otherwise} \end{cases}$	$w_1 = 2$
$f_2(state, action) = \begin{cases} 1 & \text{if } action \text{ wait is Short} \\ 0 & \text{otherwise} \end{cases}$	$w_2 = 1$
$f_3(state, action) = \begin{cases} 1 & \text{if } action \text{ speed is Fast} \\ 0 & \text{otherwise} \end{cases}$	$w_3 = 0.5$

(a)
Calculate Q('Well', 'Big Dipper'):

(b)
Apply a Q-learning update based on the sample ('Well', 'Big Dipper', 'Sick', −10.5), using a learning rate of $\alpha = 0.5$ and discount of $\gamma = 0.5$. What are the new weights?

(c) Using our approximation, are the Q-values that involve the sick state the same or different from the corresponding Q-values that involve the well state? In other words, is Q('Well', action) = Q('Sick', action) for each possible action? Why / Why not? (in just one sentence)

Now we will consider the exploration / exploitation tradeoff in this amusement park.

(d) Assume we have the original weights from the table on the previous page. What action will an ϵ-greedy approach choose from the well state? If multiple actions could be chosen, give each action and its probability.

Q2
Answer:

(a)

1 + 2 + 0 + 0.5 = 3.5

(b)

Difference = $-10.5 + 0.5 * max(4, 3.5, 2.5, 2.0, 2.0) - 3.5 = -12$
$w0 = 1 - 6 * 1 = -5$
$w1 = 2 - 6 * 1 = -4$
$w2 = 1 - 6 * 0 = 1$
$w3 = 0.5 - 6 * 1 = -5.5$

(c)

Same
They are the same because we have no features that distinguish between the two states.

(d)

With probability (1 – ϵ*4/5) we will choose the Wild Mouse. Each other action will be chosen with probability $\epsilon/5$

Q3
In this question, you will play a simplified version of blackjack where the deck is infinite and the dealer always has a fixed count of 15. The deck contains cards 2 through 10, J, Q, K, and A, each of which is equally likely to appear when a card is drawn. Each number card is worth the number of points shown on it, the cards J, Q, and K are worth 10 points, and A is worth 11. At each turn, you may either hit or stay. If you choose to hit, you receive no immediate reward and are dealt an additional card. If you stay, you receive a reward of 0 if your current point total is exactly 15, +10 if it is higher than 15 but not higher than 21, and −10 otherwise (i.e. lower than 15 or larger than 21). After taking the stay action, the game enters a terminal state end and ends. A total of 22 or higher is referred to as a bust; from a bust, you can only choose the action stay. As your state space you take the set {0, 2, . . . , 21, bust, end} indicating point totals, "bust" if your point total exceeds 21, and "end" for the end of the game.

(a)
Given the partial table of initial Q-values below, fill in the partial table of Q-values on the right after the following episode occurred. Assume a learning rate of 0.5 and a discount factor of 1. The initial portion of the episode has been omitted. Leave blank any values which Q-learning does not update.

Episode

s	a	r	s	a	r	s	a	r
19	hit	0	21	hit	0	bust	stay	-10

Initial values

s	a	$Q(s,a)$
19	hit	-2
19	stay	5
20	hit	-4
20	stay	7
21	hit	-6
21	stay	8
bust	stay	-8

Updated values

s	a	$Q(s,a)$
19	hit	
19	stay	
20	hit	
20	stay	
21	hit	
21	stay	
bust	stay	

Unhappy with your experience with basic Q-learning, you decide to featurize your Q-values

$$f_1(s,a) = \begin{cases} 0, & \text{if } a = \text{stay}; \\ +1, & \text{if } a = \text{hit and } s \geq 15; \\ -1, & \text{if } a = \text{hit and } s < 15; \end{cases} \quad \text{and} \quad f_2(s,a) = \begin{cases} 0, & \text{if } a = \text{stay}; \\ +1, & \text{if } a = \text{hit and } s \geq 18; \\ -1, & \text{if } a = \text{hit and } s < 18. \end{cases}$$

(b)
Circle all of the following partial policy tables for which it is possible to represent Q-values in the form w1*f1(s, a) + w2*f2(s, a) that imply that policy unambiguously (i.e., without having to break ties).

(i)

s	$\pi(s)$
14	hit
15	hit
16	hit
17	hit
18	hit
19	hit

(ii)

s	$\pi(s)$
14	stay
15	hit
16	hit
17	hit
18	stay
19	stay

(iii)

s	$\pi(s)$
14	hit
15	hit
16	hit
17	hit
18	stay
19	stay

(iv)

s	$\pi(s)$
14	hit
15	hit
16	hit
17	hit
18	hit
19	stay

(v)

s	$\pi(s)$
14	hit
15	hit
16	hit
17	stay
18	hit
19	stay

Q3
Answer:

(a)
Updated values:

$Q(19, hit)$ = (1 − 0. 5) * (− 2) + 0. 5 * (0 + 1 * $max(8, -6)$) =− 1 + 4 = 3
$Q(21, hit)$ = (1 − 0. 5) * (− 6) + 0. 5 * (0 + 1 * (− 8)) =− 3 − 4 =− 7
$Q(buts, stay)$ = (1 − 0. 5) * (−8) + 0. 5 * (-10 + 1 * (0)) =− 4 − 5= −9

(b)

if the policy is optimal without tie breakers, its choice according to each state and action must be the best possible.

For example in (i)
$Q(14, hit) > Q(14, stay) = 0$
$Q(14, hit) = -w1 - w2 > 0$

and also:
$Q(18, hit) = w1 + w2 > 0$

contradiction.

So in all situations that for state 14 and 18 or 19 the policy chooses hit, there is a contradiction. (eliminating policies i, iv, v)

consider (ii)

if for both 14 and 18, stay is the optimal action, we have:

$Q(18, stay) = 0 > w1 + w2 = Q(18, hit)$
$Q(14, stay) = 0 > -w1 - w2 = Q(14, hit)$

contradiction.
So the only policy that has no tie breakers is (ii)